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Cellulose acetate from palm oil bunch waste for forward osmosis membrane in desalination of brackish water

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ABSTRACT

Hygienic drinking water availability has become a significant problem in many parts of the world in emerging countries, developing countries, and the countries that often experience disasters, including Indonesia. Forward osmosis (FO) membrane is an alternative water treatment method, particularly in emergency water supplies. This study has developed a FO membrane for desalination brackish water to produce energy drinking water. The FO membrane was made from cellulose acetate (CA) polymer synthesized from cellulose isolated from waste palm oil empty fruit bunches (POEFB). The forward osmosis membranes were made by printing a polymer solution on a glass plate, followed by solidification and membrane formation. FT-IR spectrum confirmed the successful synthesis of cellulose acetate. The obtaining cellulose acetate showed that the cellulose produced was cellulose diacetate, with acetyl content and degree substitution, 36.17% and 2.15, respectively. The Cellulose diacetate FO membrane has a water flux value of 2.2 L/m^{2}hour and a rejection factor of 96% using 1 M sucrose as a draw solution. The water quality of the FO product has met drinking water standards based on the regulation of the Minister of Health of the Republic Indonesia Number 492/MENKES/PER/VI/2010 and WHO standard drinking water quality.

1. Introduction

Population growth and industrial growth have caused the need for water to continue to increase from year to year, both for domestic and industrial needs, causing various countries in the world to experience water crises [1]. The water crisis is also exacerbated by increasing pollution and environmental damage, where the available water sources cannot be used directly as a source of clean water [2]. On the other hand, the abundant availability of brackish water and sea water cannot be directly used because they contain high salinity, so unique treatment processes are required, which tend to be expensive. Membrane technology has shown great potential and is effective for separating and obtaining clean water and drinking water to overcome the problems [3]. The membrane technology commonly used for processing and purifying brackish water and seawater into clean water is the reverse osmosis (RO) membrane [3]. The general advantages of membrane technology are that it is practical and straightforward, does not require additional chemicals, is easily regenerated, and has good chemical resistance [4,5]. However, RO membranes also have several weaknesses, including being considered energy-intensive and expensive and environmental problems.

In recent years, forward osmosis (FO) has become a center of attention for researchers as an alternative method for desalination and purification of seawater, brackish water, and dirty water. FO membranes have advantages, including lower energy consumption and a very economical process compared to conventional technology [6]. FO is a water purification process that is driven by the osmotic pressure generated by a high concentration attractant solution to separate water from unwanted solutes using a semipermeable membrane [7]. Two primary keys determine the performance of the FO membrane, namely the membrane material and the draw solution [8]. Many researchers have developed and applied various draw solutions to obtain high water fluxes and improve the performance of FO membranes [9–12], including ways to minimize membranes [13]. The FO membrane material used can be in the form of natural polymer materials or synthetic polymers. Saiful et al. [5] have made FO membranes from chitosan and chitosan-starch composites as natural polymers that can be used to purify brackish water, seawater and dirty water. Researchers are continuously trying to find cheaper and environmentally friendly FO membrane materials by utilizing natural polymers whose raw materials are abundantly

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available. Besides chitosan and starch, another natural polymer source that is abundantly available and interesting to develop is cellulose acetate which can be synthesized from cellulose biomass. Cellulose is a naturally occurring, renewable, and abundantly available macromolecular polysaccharide in nature both in wood and agricultural crops, biomass waste, wood and plant residues, and other organic waste materials [14]. The development and application of cellulose continue to interest researchers because it is abundantly available and easily renewable and has unique properties such as hydrophilicity, chirality, and biodegradability [15]. One of the available waste biomass as a source of cellulose is the palm oil empty fruit bunches (POEFB) waste. The POEFB has considerable potential to be used as a source of cellulose because the POEFB has a high cellulose content of around 65% compared to the amount of cellulose obtained from other agricultural wastes [16]. The total production of oil palm plantations in Indonesia, in particular, can reach 33 million tons and is increasing yearly. Oil palm plantations produce 23% of the POEFB waste per tonne of oil palm bunches [17]. Therefore, it is necessary to exploit the potential of these local oil palm empty fruit bunches as an effort to deal with palm oil waste and be used as other more valuable materials such as membranes.

In this study, the manufacture of FO membranes made from cellulose acetate polymer was synthesized from isolated cellulose contained in the POEFB. Isolation of cellulose from the POEFB through the stages of the process of alkalization, acid hydrolysis, and bleaching. Cellulose acetate membranes were prepared by the phase inversion method. The obtained membranes were characterized for membrane structure, swelling degree, porosity, mechanical properties, and water flux. The FO membranes were applied for the desalination of brackish water. Sucrose is used as a draw solution to produce drinking water containing calories that can be used in an emergency. The quality of the water produced was evaluated on several parameters, including salinity, pH, conductivity, and total dissolved solid.

2. Experimental

2.1. Equipment and chemicals

The equipment used in this study were glassware, oven, analytical balance, hot plate stirrer, stirring rod, a Buchner funnel accompanied by a glass flask and vacuum, membrane module, pH meter, stopwatch, knife, micrometer digital Mitutoyo, tensile strength (Universal Testing Machine), SEM (Hitachi SU 3500), FT-IR (Shimadzu Prestige).

Palm oil empty fruit bunches were obtained from PT. KTS, West Aceh and brackish water is taken from Krueng Cut, Aceh Besar. All chemicals used are chemicals with pro-analysis quality. The chemicals used were glacial acetic acid (Merck, Darmstadt, Germany), anhydrous acetic acid (Merck, Darmstadt, Germany), sulfuric acid 97% (Merck, Darmstadt, Germany), hydrochloric acid 36% (Merck, Darmstadt, Germany), distilled water, polyvinylpyrrolidone, methanol (Merck, Darmstadt, Germany), 1,4-dioxane (Merck, Darmstadt, Germany), acetone (Merck, Darmstadt, Germany), lactic acid (Merck, Darmstadt, Germany), sodium hypochlorite 15% (Jeungkimali, Banda Aceh, Indonesia), sodium hydroxide (Merck, Darmstadt, Germany), toluene (Merck, Darmstadt, Germany), ethanol (Merck, Darmstadt, Germany), sodium acetate (Merck, Darmstadt, Germany) and filter paper.

2.2. Experimental procedures

2.2.1. Celulluse isolation

Cellulose isolation from oil palm empty fruit bunches was carried out through a three-step process as described below.

2.2.1.1. Alkalization. Alkalization process for waste the POEFB refers to Abral et al. [18] with modification. Isolation of cellulose was preceded by the preparation of samples of waste from the POEFB, which were washed and soaked in hot water. Then the fiber is cut into small pieces and rinsed with water to remove the remaining impurities. After that, air-dried and blended until the fibers are smooth. Therefore, 20 g of oil the POEFB are soaked in a mixture of toluene and ethanol (2:1) for 24 h. Then the fiber was dewaxed using a hot plate stirrer for 48 h, 500 rpm at 50 °C, then rinsed with distilled water until it reached pH 7 accompanied by filtration using a Buchner funnel accompanied by a glass flask and vacuum system. The residue was then dried in an oven for 20 h at 50 °C. Furthermore, the dry fiber was soaked in 5% NaOH and re-dewaxed using a hot plate stirrer for 4 h, 500 rpm at a temperature of 50 °C, then rinsed with distilled water to pH seven accompanied by filtering using a Buchner funnel system. The residue was dried in an oven for 20 h at 50 °C, then mashed using a blender to obtain fine fiber.

2.2.1.2. Acid hydrolysis. The fine fiber was then hydrolyzed using 5 M HCl for 12 h at 50 °C, then rinsed using distilled water until it reached pH 7 accompanied by filtration using a vacuum lined with filter paper.

2.2.1.3. Bleaching. The soft fiber is bleached to remove the remaining lignin that is still in the fiber. The fiber was added with a mixture of 15% NaOCl and CH_3COOH (4:1) and then stirred using a hot plate stirrer for 2 h, 500 rpm at 60 °C. Then rinsed using distilled water to pH 7 and filtration using vacuum suction, a white powder was obtained. The cellulose powder was then dried in an oven at 40 °C and characterized by FTIR and SEM.

2.2.2. Synthesis cellulose acetate

The synthesis of cellulose acetate begins with the activation step; by mixing 5 g of cellulose and 50 mL of glacial acetic acid, stirred with a hot plate stirrer at 38 C and 300 rpm for 3 h. In the next stage, the acetylation stage, adding 15 mL of acetic anhydride and 16 drops of concentrated sulfuric acid as a catalyst to stir again at 30 C for 2.5 h until the solution turns brownish-yellow. Then added 2 mL of distilled water and 5 mL of glacial acetic acid, and stirred for 30 min. The last stage is hydrolysis by adding 2 mL of distilled water and 5 mL of glacial acetic acid with a reaction time of 30 min. Then add 1 g of sodium acetate and wait for 5 min. The dark-colored solution formed was then poured into distilled water and stirred vigorously. The white precipitate of cellulose acetate formed was then filtered and rinsed until the odor of acetic acid disappeared. The cellulose acetate precipitate was then dried in an oven at 50 C for 6 h and characterized using FTIR and SEM.

2.2.3. Preparation of cellulose acetate membrane

Cellulose acetate membranes were prepared as described by Shang & Shi [19] and Mirkhalili et al. [20] with slight modifications. Cellulose acetate membranes were prepared by the phase inversion method. Membrane dope solution was prepared from a mixture of cellulose acetate 13% (wt), 1,4 dioxane 22.5% (wt), acetone 15.5% (wt), polyvinylpyrrolidone 1.5% (wt), lactic acid 1% (wt).) and 3% methanol (by weight). The mixture was stirred using a magnetic stirrer for ±8 h to form a homogeneous mixture. The dope solution was then printed on a glass plate after 24 h. The impression solution was allowed to stand at room temperature for 30 s and then put into a coagulation bath containing water. The formed membrane is stored in ion-free water before further use. The resulting membrane was further characterized and used for performance evaluation using the FO module.

2.2.4. Membrane characterizations

2.2.4.1. FT-IR analysis. Cellulose and cellulose acetate powder as much as 1 g were characterized by chemical structure using Spectroscopy Fourier Transform Infra-Red (FTIR - Shimadzu) at wave number 4000–500 cm-1. This test was carried out at the Instrumentation Laboratory of the USK Chemical Engineering Department.

2.2.4.2. SEM analysis. Cellulose powder, cellulose acetate and cellulose acetate membranes were characterized using SEM to view their surfaces with 250x and 1500x magnifications, and cellulose acetate membranes (cut to 2 × 1 cm in size) were characterized using SEM (Hitachi SU 3500) to view the top, bottom and transverse surfaces with this magnification. 1500x. This test was carried out at the FMIPA Laboratory, Bandung Institute of Technology and LIPI Serpong.

2.2.4.3. Swelling degree and membrane porosity analysis. The swelling degree of the membrane was determined by weighing the wet and dry membranes three times each. Then the value is determined using equation (1).

$$sd(\%) = \frac{W_{wet} - W_{dry}}{W_{dry}} \times 100 \quad (1)$$

where W_{wet} is the weight of the wet membrane and W_{dry} is the weight of the dry membrane.

Therefore, the porosity test measured the diameter and thickness of the membrane using a digital micrometer. Porosity can be determined using equation (2).

$$\varepsilon(\%) = \frac{V_{wet} - V_{dry}}{V_{dry}} \times 100 \quad (2)$$

where ε (%) is the porosity of the membrane, V of the wet membrane is the total volume of the wet membrane, and V of the dry membrane is the total volume of the dry membrane.

2.2.4.4. Contact angle measurement. The contact angle is measured by dripping water from a height of 2 cm above the membrane surface using a syringe. Water droplets on the substrate surface were photographed from the front using a DSLR camera with adjusted contrast, light, and focus settings. Furthermore, the images are processed using ImageJ software.

2.2.5. FO experiment

The Cellulose acetate FO membranes were evaluated using the FO module with a continuous flow system. The FO module is illustrated in Fig. 1, which consists of two compartments, the brackish water feed flow compartment and the sucrose as draw solution flow compartment. The FO process was carried out with variations in the concentration of the sucrose as draw solution; 0.5 M, 1 M, 1.5 M, and 2 M for 1 h FO process. Furthermore, the feed and draw solution parameters were evaluated for pH, salinity, TDS, and conductivity.

The water flux from the FO membrane was determined for 6 h of FO filtration. The flux value obtained is calculated using equation (1)

$$J = \frac{V}{\pi r^2 t} \quad (3)$$

where J is the water flux (L/m^{2}h), r is the membrane radius (m), V is the permeate volume (L), and T is the time (hours).

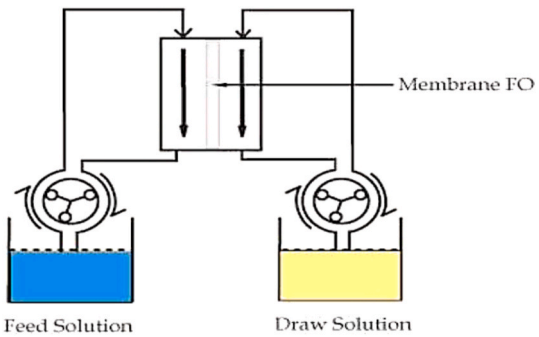


Fig. 1. Illustration of the FO module used in this study.

Therefore, the membrane rejection factor is determined by based on the rejection total of dissolved components in the feed solution by the cellulose acetate membrane. The rejection factor (R) was calculated using equation (2):

$$R\ (\%) = 1 - \frac{Cp}{Cf} \times 100\% \quad (4)$$

where R is the percentage of salt rejection, Cp is the solute/salt concentration in the permeate (TDS), and Cf is the solute/salt concentration in the feed (TDS).

3. Results and discussion

3.1. Characterization of synthesized cellulose and cellulose acetate extract

Cellulose has been isolated from oil palm empty fruit bunches by a three-step process: alkalization, acid hydrolysis, and bleaching. Meanwhile, the synthesis of cellulose acetate from cellulose also involves three stages; activation, acetylation, and hydrolysis. Fig. 2 shows the SEM pictures for cellulose from POEFB 2(a) and cellulose acetate from its synthesis 2(b). SEM images show that particles cellulose has a less uniform morphology and porous structure, while Fig. 2 (b) of cellulose acetate shows morphological changes, which are more porous and more uniform.

The success of cellulose isolation and cellulose acetylation reaction can be proven from the identification using FTIR, which shows the change in the functional group from cellulose to cellulose acetate after forming acetyl groups. The results of the FTIR analysis of cellulose and the change in functional groups from cellulose to cellulose acetate are shown in Fig. 3.

Fig. 3 (a) is cellulose that can be seen from a wavelength of 3400 cm^{-1} caused by stretching of the –OH group [21], while cellulose from POEFB observed at a wave number of 3348 cm^{-1} caused by the replacement of free OH groups for acetyl groups [22] and showed the presence of intramolecular hydrogen bonds in cellulose. The absorption at 2900 cm^{-1} indicated the presence of a CH_2 group [23]. Absorption at a wavelength of 1640 cm^{-1} indicates that there is an interaction between cellulose and water [24]. Then the absorption of 1424 cm^{-1} indicates the presence of CH_2 bonds. Absorption at a wave number of 1073 cm^{-1} indicates the C–O group of alcohol on the cellulose group [25].

The synthesis of cellulose acetate from POEFB has been successfully carried out, which can be seen in Fig. 3(b). The FT-IR spectrum of commercial cellulose acetate is also shown (Fig. 3(c)). Based on this comparison, it can be seen that the functional group absorption area of cellulose acetate from OPEFB resembles that of commercial cellulose acetate. The typical absorption peak for cellulose acetate is the absorption peak of the C═O carbonyl group and the C–O ester group of the acetyl group, where the absorption peak of the carbonyl group on cellulose acetate from OPEFB is located in the 1748 cm^{-1} region while commercial cellulose acetate is at 1743 cm^{-1}. This indicates that the OH functional group in cellulose has been substituted by a C═O carbonyl group [25] in the acetylation process. The C–O ester group in the cellulose acetate spectrum of OPEFB is in the 1230 cm-1 region, whereas in commercial cellulose acetate, it is in the 1224 cm^{-1} region. The absorption of other functional groups is at wave number 2954 cm^{-1} which indicates stretching C–H vibration and wave number 1363 cm^{-1} indicates C–H bending vibration of the acetyl group [26], while for cellulose acetate commercially seen at wave numbers 2945 cm^{-1} and 1363 cm^{-1}. Finally, the O–H hydroxyl group in OPEFB cellulose acetate has a relatively higher absorption seen at a wave number of 3620 cm^{-1} compared to commercial cellulose acetate.

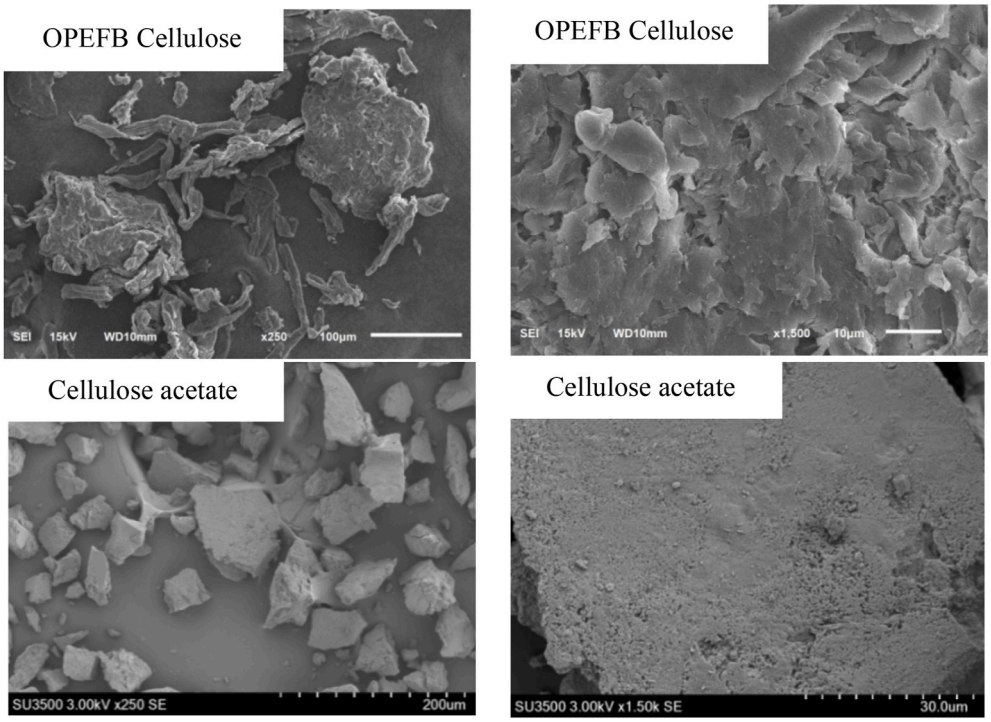


Fig. 2. SEM results of OPEFB cellulose (a) and cellulose acetate (b) with different magnification (i) 250x and (i) 1500x

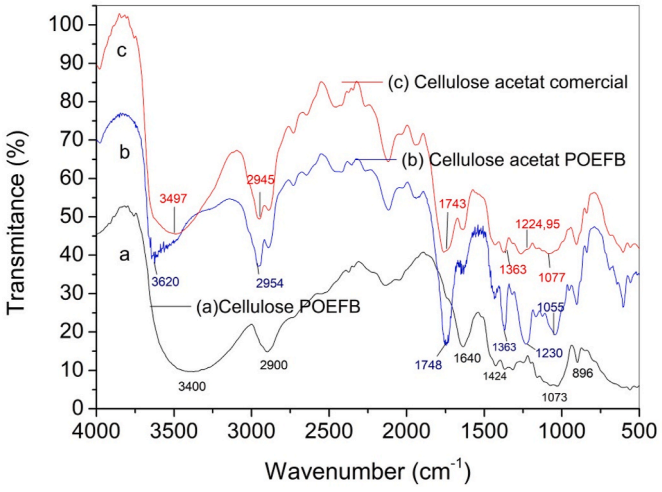


Fig. 3. FT-IR spectrum of test results from (a) cellulose, (b) POEFB cellulose acetate, and (c) commercial cellulose acetate.

3.2. The cellulose acetate membrane preparation and characterization

POEFB cellulose acetate membrane was prepared using the phase inversion method. The composition of the dope solution and the phase inversion process affect the formed membrane's structure and morphology. Visually, the cellulose acetate membrane obtained is a thin membrane with a tight and flat surface structure with a thickness of 0.145 mm. Fig. 4 shows the SEM images of the cellulose acetate membrane on the top, bottom, and cross-section. The morphology of the membrane shows differences in morphological structure. The cellulose acetate membrane formed has an asymmetrical structure, where the top surface of the membrane is tight but less evenly distributed, while the bottom surface is more dense and homogeneous. Based on Fig. 4(a), shows the top surface of the cellulose acetate membrane which is tight but uneven. While the structure of the lower surface of the membrane 4 (b) is denser and homogeneous, this is also supported by Fig. 4(c) both sides of the top and bottom layers are tight membranes, where the bottom layer is denser and thicker, there is no finger like macrovoid on the inner membrane. The cross-section of membrane 4(c) shows an asymmetrical and porous structure [27]. The pores of the cellulose acetate membrane are influenced by the percentage of cellulose acetate and the ratio of the solvent and its additives, in which case the dope solution is thick. The thicker the dope solution, the smaller the resulting

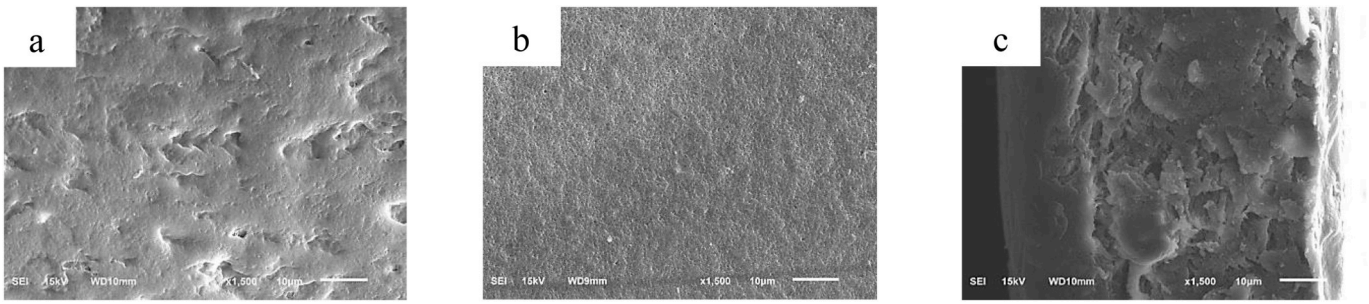


Fig. 4. Morphology of cellulose acetate membrane with 1,500x magnification in (a) top surface, (b) bottom surface, and (c) cross-section.

pores [20]. The intense interaction between solvent and non-solvent results in a membrane with a denser structure [26].

The POEFB cellulose acetate membrane had a porosity of 15%, with a swelling degree of 39%. The swelling degree and porosity of the membrane are closely related to the sensitivity of the membrane to water and describe the percentage of the porous membrane structure [26]. Then the cellulose acetate membrane from OPEFB has relatively good mechanical properties with tensile strength and elongation values obtained are 35 MPa and 36%, respectively. The microstructure and intermolecular forces influence the high mechanical properties of this membrane along with the cellulose acetate membrane structure [28]. Cellulose acetate POEFB is a type of diacetate which has greater intermolecular forces. The contract angle of the OPEFB cellulose acetate membrane was 66^0, indicating hydrophilic membrane properties.

3.3. The cellulose acetate membrane FO application

The OPEFB cellulose acetate membranes were investigated for brackish water desalination. The desalination was carried out using the FO module to determine the cellulose acetate membrane's water flux and percent salt rejection with a brackish water feed solution. The cellulose acetate membrane had a surface area of 0.036 m^2 for 6 h. This test uses 1 M sucrose as the draw solution, producing a high water flux and rejection percentage in brackish water purification [5]. Fig. 5(a) informs that the highest water flux value was obtained at an operating time of 1 h and the water flux decreased with increasing time. The water flux produced by the cellulose acetate membrane for 1 h is 2.22 L/m^2h. A decrease in the value of water flux occurred during the process, precisely after 1 h, correlated to a decrease in the osmotic pressure in the sucrose solution due to dilution by water molecules moving into the draw solution and also allegedly due to the presence of solutes in brackish water accumulating in the active layer of the membrane which resulted in the rate of water movement is hampered [29].

OPEFB Cellulose acetate membranes have a high rejection percentage of solute components contained in brackish water, although there is a slight decrease during the FO process. Fig. 5(b) informs that the cellulose acetate membrane decreased the rejection value with increasing operating time. The highest rejection percentage was obtained for 1 h at 96%. Thus, the cellulose acetate membrane of OPEFB can separate total dissolved solute in brackish water relatively high. Based on the flux value and the rejection percentage concerning the time at a concentration of 1 M sucrose solution, the optimum FO process for the OPEFB cellulose acetate membrane is 1 h.

The value of the water flux in the FO process using a cellulose acetate membrane is influenced by the concentration of sucrose as a draw solution. Fig. 6(a) demonstrates the effect of sucrose concentration on the water flux produced during the 1-h FO process. An increase in the sucrose concentration increases the osmotic pressure, thereby increasing the amount of water to move into the draw solution side. The concentration of sucrose solution forms a linear relationship with the water flux, where the higher the concentration, the greater the water flux was obtained. These results indicate that the sucrose concentration can be selected according to the expected water flux value. At the same time, Fig. 6(b) shows the rejection percentage in variations of the concentration of sucrose solution. The total solute rejection percentage was not influenced by increasing the concentration of the sucrose solution. Based on this, the resulting rejection percentage is above 95%, meaning the separation occurs relatively well.

3.4. Quality of water FO product

In the FO process, sucrose was used as a draw solution to produce drinking water containing sugar and become a high-calorie energy drink. These energy drink water products are suitable for water needs in emergencies. The water quality of FO products was analyzed against several parameters related to the quality of brackish water and standard drinking water quality, including pH, salinity, total dissolved solids, and conductivity. Brackish water as a feed solution has a pH of 6.5, salinity of 9.5, and total dissolved solids of 1245 mg/L. The Effect of sucrose concentration as draw solution to FO product water on pH parameter is shown in Fig. 7. The pH of FO water product using a cellulose acetate membrane experienced insignificant changes as the concentration of the sucrose solution increased. The initial pH of brackish water before the FO process was 6.5 and increased to 7.1 when the sucrose concentration was 2 M. The pH water FO product obtained met the regulation of the Minister of Health of the Republic of Indonesia for drinking water standards and WHO standard water quality, in which the pH range of the water that can be consumed is 6.5–8.5.

The results of the salinity of the brackish water feed solution and the FO product water are shown in Fig. 8. Fig. 8 shows a minimal salinity change in brackish water and FO water products. The initially measured salinity of brackish water was 9.5 ppt and decreased to 9.2 ppt as the draw solution concentration increased. The cellulose acetate membrane is selective and able to retain dissolved salts in brackish water so that it does not pass through the membrane. The salinity of the sucrose solution also proves that the results during and after the FO process remain unchanged. A similar incident was also experienced by Saiful et al. [30], both the operation of the FO membrane with variations in the concentration of the draw solution to brackish water and sea water [5], where the resulting salinity is 0–1.3 ppt as the concentration of the draw solution increases. This salt content indicates that the water is still classified as fresh water and is below the maximum salinity limit for commonly used water, which is five ppt. The salinity indicates that the FO cellulose acetate membrane works well as the membrane filter for the desalination brackish water to produce drinking water.

The performance capability of the FO membrane can also be

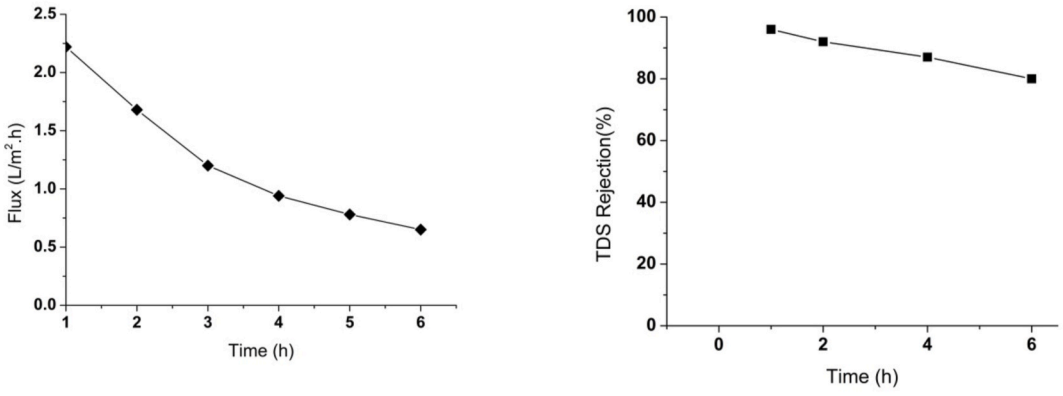


Fig. 5. Water flux (a) and percent rejection (b) of a cellulose acetate membrane using a 1 M sucrose as draw solution and brackish water as feed solution.

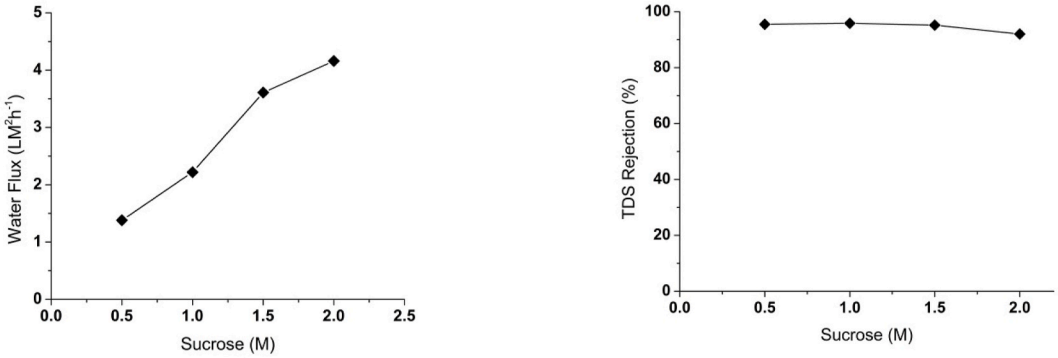


Fig. 6. Effect of sucrose concentration as draw solution on (a) water flux and (b) rejection percentage of total dissolved solute.

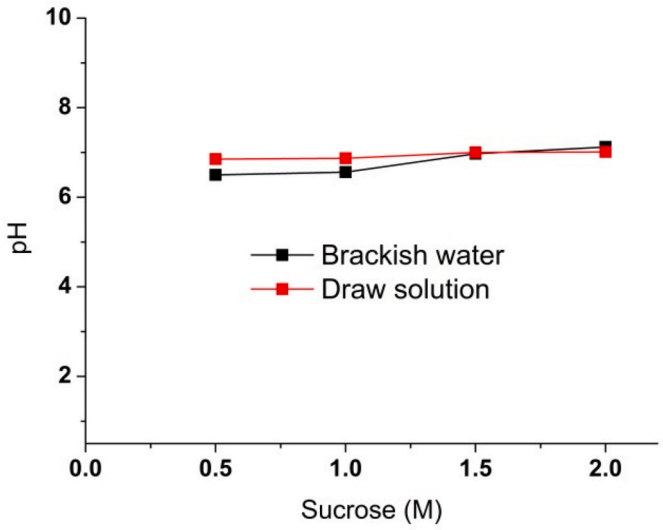


Fig. 7. Effect of sucrose concentration as draw solution on the pH of FO water product.

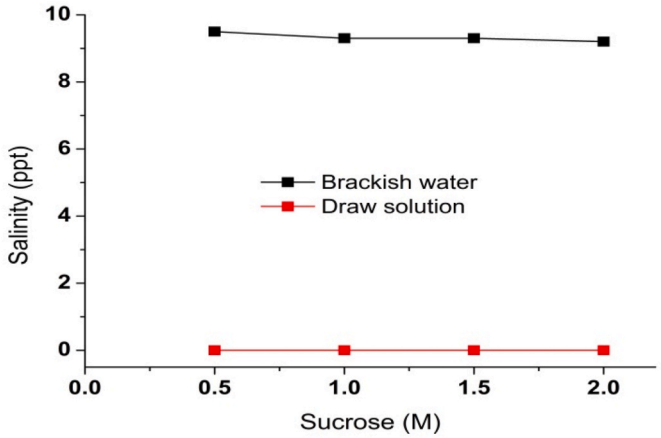


Fig. 8. Effect of sucrose concentration as draw solution on product water salinity.

observed from the TDS analysis, which can also be related to the rejection percentage. The change in TDS of the brackish water feed solution and FO product water can be seen in Fig. 9(a). Fig. 9 informs that the results of the TDS analysis of brackish water experienced a slight decrease and FO product water experienced a slight change as the concentration of sucrose solution increased, especially at a concentration of 2 M. The TDS of brackish water at a concentration of 0.5 M draw solution was 1245 mg/L and decreased to 1164 mg/L at a concentration of 2 M draw solution. TDS in water the product changed from 56 mg/L at a concentration of 0.5 M to93 mg/L at a concentration of 2 M sucrose solution. These results are also similar to the previous result [4], where the TDS in the draw solution increases with increasing concentration, namely at a concentration of 2 M at 401 mg/L. However, these results do not exceed the TDS limit allowed for drinking water standards by the Minister of Health of the Government of the Republic of Indonesia and WHO drinking water quality, which is 500 mg/L. It can be concluded that the cellulose acetate membrane can be used in the purification process of brackish water because it is able to hold total dissolved substances in brackish water.

The slight change in the TDS value is also supported by the water conductivity analysis of the feed solution and the sucrose draw solution. The results of this analysis can be seen in Fig. 9(b) shows that the conductivity in brackish water experienced a slight decrease from its initial condition as the concentration of the sucrose solution increased. The initial conductivity value of brackish water was 16.43 mS/cm and changed to 15.64 mS/cm after 1 h. At the same time, the FO product water slightly experienced a change in conductivity as the concentration of the sucrose solution increased. These changes range from 0.13 mS/cm to 0.22 mS/cm at a concentration of 2 M. However, the displacement of these electrolyte components can still be classified as very low.

In the FO process, the flux values are strongly influenced by the concentration of the draw solution. The draw solution concentration will continue to decrease during the FO process as the volume of water from the feed solution moves to the draw solution. The decreasing concentration of the draw solution causes the water flux value to continue to decrease during the FO process until it reaches a state of minimum flux value. However, the FO process which was carried out for 6 h using 1 M sucrose solution as a draw solution, still had a water flux value. The observations found that it had a maximum water flux value in the first hour. The water flux value obtained is still comparable to commercial FO membranes. However, the value of the OPEFB membrane water flux can still be increased by increasing the draw solution concentration as shown in Fig. 6(a) and increasing the porosity and hydrophilicity of the OPEFB membrane. When the sucrose concentration was increased to 2 M, the water flux increased to 4.2 $Lm^{-2}h^{-1}$. The resulting water flux has a relatively good value compared to commercial membranes for application to actual saline water samples, as shown in Table 1.

The cellulose acetate POEFB membranes have good performance compared to commercial FO membranes, as shown in Table 1. The cellulose acetate POEFB membranes in their application for water desalination, especially in brackish water, have water flux values close to CTA membranes and even higher than commercial CTA membranes depending on the type of draw solution used in the CTA membrane FO test. The POEFB cellulose acetate membrane has a water flux value of 4.2 $Lm^{-2}h^{-1}$ above the CTA and TFC-PS (HTI) membranes. The Cellulose

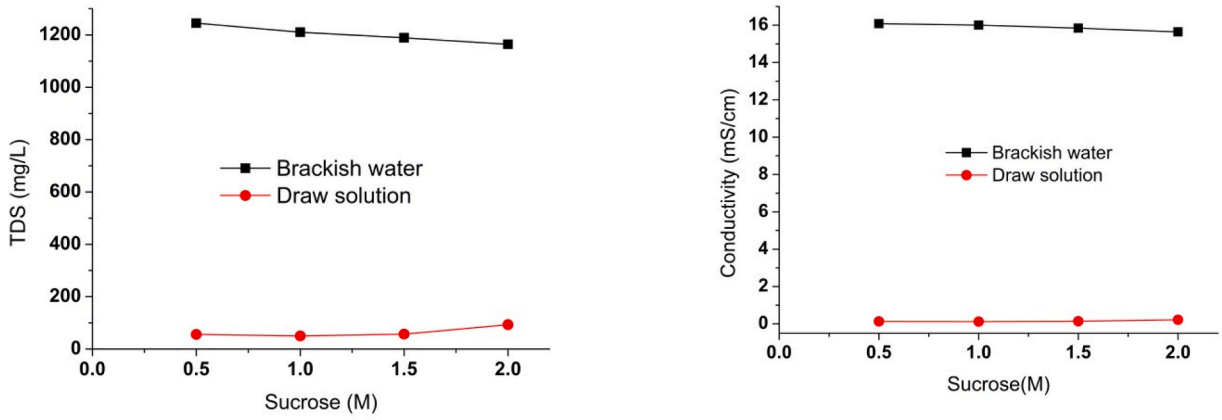


Fig. 9. Effect of sucrose concentration as draw solution on (a) TDS and (b) conductivity.

Table 1
The main characteristics of the developed FO membrane compared with other similar published works highlight the advantages of the developed FO membrane using actual process water.

Membrane	CTA, TFC	CTA	CTA, TFC	PEI-HDFA	TFC-PS	CA POEFB
Source	HTI	HTI	HTI	Non-commercial	HTI	Non-commercial
FO Application	Produced water	High salinity produced water	Industrial water	1.1 M NaCl DI water	Brackish water	Brackish water
Draw solution	NaCl	NH3–CO2 6 M, MgCl2 4.8 M	Synthetic seawater	gelatin-coated MNPs	CO(NH2)2, 10 g/L NaCl, 10 g/L	Sucrose 2 M
Water flux (L/m2h)	7,9 (CTA) 11 (TFC)	1 (NH3–CO2), 4.5 (MgCl2)	1.7 (CTA) 9.9 (TFC)	0.62 1.54	1.378–2.234 (CO(NH2)2) 7–7.96 (NaCl)	4.2
Salt rejection R (%)	NA	NA	NA	NA	84.6–89.8 (CO(NH2)2, 96.44–96.86 (NaCl)	96
Reference	[31]	[32]	[33]	[34]	[35]	In this research

acetate POEFB membrane also has a high salt rejection ability and is similar to the ability of the TFC-PS membrane; the salt rejection ability of the POEFB membrane is 96% equivalent to the ability of the TFC-PS membrane.

4. Conclusions

Cellulose acetate membrane from OPEFB has been successfully synthesized by producing a FO membrane with a tight and flat structure, hydrophilic, and high mechanical properties. OPEFB cellulose membrane has a water flux of 2.22 L/m^{2}h and a rejection percentage of 96% using 1 M sucrose solution as a draw solution. The water quality of FO products meets the drinking water quality standards according to the Minister of Health of the Government of the Republic of Indonesia and the WHO drinking water quality standards for parameters of pH, salinity, TDS, and conductivity. FO product water can be used as alternative drinking water in an emergency, which contains sugar as a calorie source.

Author contributions

Conceptualization, methodology, formal analysis, data curation, supervision, funding acquisition, S.S., S.H., N.K.; resources, project administration, validation, writing — original draft preparation, S.S., S. H.; writing —review and editing, R.R., S.S. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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